Project Eco: Mid-Submission Report (M2)

Green Home Score and retrofit recommendation engine

STATUS AT A GLANCE

The scoring, energy, carbon, planning, value and explanation engines are built, tested (32 of 32 tests pass) and measured on seven experiments. Five of six M1 test targets are met; value-range coverage (77.4% against 80%) is not yet. The dashboard and the live language model are still to build; the plan on page 2 sets out how they will be finished.

1. Current approach and changes since M1

Project Eco answers three questions for a homeowner or mortgage advisor: how green a home is today, which improvements suit it, and what those improvements would save in energy and carbon and add in value. The structure set out in M1 is unchanged: a validated property record passes through five components (Figure 1). The governing rule is that every figure shown to a user is produced by a calculation or a trained model, and the language model only describes results it is given.

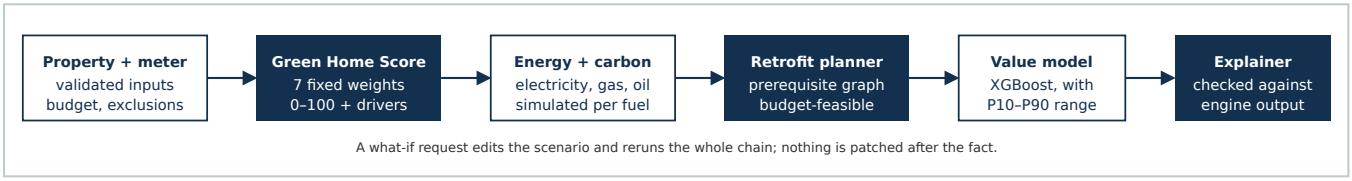


Figure 1. Pipeline as implemented.

CHANGES SINCE M1

1. **Value lift is now isolated to the rating channel.** Our first what-if swapped every retrofitted feature into the value model at once and produced a negative value change for 129 of 300 retrofitted homes, which is not credible. The rating now enters as a single ordinal feature with a monotonic constraint, so a better rating cannot lower the predicted value. Negative cases fell to 0 of 300 with held-out accuracy unchanged.
2. **The planner searches exhaustively.** The catalogue has five measures, so there are only 31 candidate packages. Instead of a path heuristic, we evaluate every package that satisfies prerequisites and budget and return the one with the largest carbon saving. This takes 0.15 ms per home.
3. **Prerequisites are explicit.** Insulation and smart controls precede a heat pump; insulation precedes solar. Experiment E4 confirms why: insulating first lowers the heat pump's electricity demand by 15%.
4. **Missing meter data is handled deliberately.** In our data 10.7% of months and 3% of homes have no reading. A missing value stays missing and is never read as zero; with fewer than six months of readings the score falls back to modelled energy intensity and re-weights the remaining components.
5. **The explanation layer is template-first.** A validated template answer is always available. A language model may be plugged in behind a validator that rejects any number not present in the engine output, so a failure or a response slower than 1.5 s never blocks a result.

2. Experiments run so far and their results

All experiments use a seeded synthetic dataset of 5,000 homes (60,000 monthly meter rows, six regions, four property types), split 80/10/10 by property, with preprocessing fitted on the training set only. Results below are on the 500 held-out homes unless stated. The dataset is synthetic because licensed EPC and sales data were not yet available to us; the generator and its label rules are documented and reproducible.

#	EXPERIMENT	SETUP	RESULT
E1	Value model against baselines	XGBoost vs. median £/m^2 by region vs. linear regression	MAE £32.3k vs. £41.8k and £66.1k; MAPE 8.4%; P10–P90 interval covers 77.4% of true values (target 80%)
E2	Score monotonicity	Raise one component one level in every test home	2,256 of 2,256 checks leave the score equal or higher
E3	Planner validity and gain	500 homes, random budgets of £5k–£35k	500/500 respect prerequisites, budget and completed work; median cut 20.5% of purchased energy, 526 kg CO_2e a year
E4	Heat-pump fuel shift	95 m^2 semi-detached, old gas boiler	Gas falls 19,143 kWh, electricity rises 4,467 kWh (a negative saving); with insulation first the rise is 3,327 kWh
E5	Value lift	300 planned homes, rating change only	118 homes improve a band; median lift 4.9% (about £15.8k); 90th percentile 7.2%
E6	Grounding guard	200 template answers, 200 injected wrong figures	200/200 template answers grounded; 195/200 wrong figures blocked
E7	Service latency	200 full scenario requests	Median 3.0 ms, 95th percentile 3.8 ms

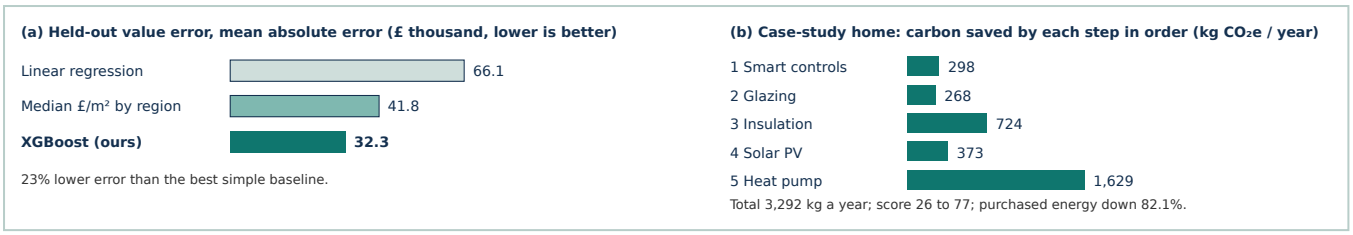


Figure 2. (a) Experiment E1. (b) Experiment E4, with a £35k budget. The heat pump saves most only because the fabric steps before it cut its electricity demand by 15%.

3. Progress on testing

All 32 automated tests pass.

AREA	PASSING	WHAT IS CHECKED
Score engine	11 / 11	Weights sum to one; score stays within 0–100; each component's mapping is monotonic; contributions add up to the rounded score; missing meter data re-weights correctly
Energy and carbon	5 / 5	Gas falls and electricity rises under a heat pump; carbon units are kg CO_2e; a zero baseline reports "unavailable"; missing readings are not zeros
Retrofit planner	8 / 8	Graph is acyclic; prerequisites precede dependents; budget is never exceeded; completed, excluded and infeasible measures (e.g. rooftop solar on a flat) are skipped; step savings sum to the total
Service and explainer	8 / 8	Invalid area or rating band is rejected; a what-if rerun gives consistent results; identical input reloads identically; an invented figure or a timeout falls back to the template

TARGET	STATUS	EVIDENCE
Groundedness at least 90%	Met	100% of template answers grounded (E6)
Unsupported claims below 5%	Met	2.5% of injected wrong figures passed the guard (E6)
Beat a simple baseline on held-out error	Met	MAE 23% lower than the median £/m² baseline (E1)
Value-range coverage near 80%	Not yet met	77.4% on held-out homes (E1)
Prerequisite ordering and budget limits	Met	500/500 plans valid (E3)
Measured response time	Met	95th percentile 3.8 ms, before database and model-call overhead (E7)

KNOWN ISSUES

- Five of 200 injected wrong figures slipped through because they matched a genuine number by chance. The guard compares bare numbers; it should compare a number together with its unit (kg, kWh, £, %).
- The value interval under-covers at 77.4%. We intend to calibrate it on the validation split using conformal prediction.
- All results are on synthetic data. They show that the engines behave correctly and consistently, not how accurate they are on real UK homes; that claim needs real EPC and sales data.

4. Plan for the remaining time

Built and tested: the scoring, energy, carbon, planning, value and explanation components and the service layer, plus the database schema. Not yet built: dashboard, sign-in, database persistence and the live language model. The steps below show how each will be completed.

#	WORK	DETAIL	DONE WHEN
1	Close known issues	Unit-aware grounding guard; conformal calibration of the value range; save scenarios to PostgreSQL and reload them unchanged	Coverage ≥ 80%; guard blocks all injected cases; save/reload test passes
2	Dashboard	Property form, score breakdown with each component's contribution, ordered roadmap, before/after by fuel, scenario comparison	One home assessed and compared end to end in the interface
3	Language model and what-if	Connect the model behind the validator; parse what-if requests into scenario changes and rerun the engines	Groundedness ≥ 90% on a fixed question set; fallback works on timeout
4	Final validation	Rerun every experiment; add a public EPC sample alongside the synthetic data; prepare the demonstration on the case-study home	Results reproduced from one command; demo runs without edits

Risks. The dashboard is the largest remaining item; the engines already return complete results through the service, so it can be built against them without further changes. Language-model failure is covered by the template answer. Data realism is the main residual risk, so real public data is a fixed item in step 4.